Tora Matsuda

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Summary

CS graduate from UC Santa Cruz with research experience in autonomous driving systems, LLM security applications, and distributed systems. Fine-tuned LLMs for MITRE ATT&CK command generation and built several ML-powered full-stack applications. Seeking full-time roles in software engineering or cybersecurity.

Education

University of California, Santa Cruz — Santa Cruz, CA
Bachelors in Computer Science — *Sep 2022 – Jun 2026*

- Relevant Coursework: Computer Security (CSE 132), Cryptography (CSE 108), Computer Networks (CSE 150), Machine Learning (CSE 144), Distributed Systems (CSE 138), Data Structures & Algorithms (CSE 101)

Experience

AIEA Lab, UC Santa Cruz — Santa Cruz, CA
Undergraduate Research Assistant — *Jan 2026 – Apr 2026*

- Benchmarked reinforcement learning algorithms (PPO, DDPG, SAC) across CARLA and OpenAI Gymnasium simulators
- Configured containerized pipelines on Kubernetes and analyzed convergence metrics to guide algorithm selection

UCSC Baskin Engineering — Santa Cruz, CA
CSE 101 Reader/Tutor (Data Structures & Algorithms) — *Jan 2026 – Apr 2026*

- Graded 500+ submissions across biweekly assessments for 100+ students; proctored exams and fielded questions on algorithm design and complexity

Projects

TasteMate – Music Social Platform | *PyTorch, FastAPI, CLAP, Spotify API* — Mar 2026 – Present

- Built a music social platform matching listeners via shared sonic identity and live listening presence
- Architected an ML engine using CLAP audio embeddings, WebSockets, and a PyTorch retraining pipeline
- Integrated the Spotify API across a 100M+ track catalog and shipped an iOS client with Capacitor

100-Class Image Classification via Transfer Learning | *Python, PyTorch, DINOv2* — Apr 2026 – Jun 2026

- Built a 100-class image classifier for a Kaggle competition on ∼1,000 training images
- Switched backbone to DINOv2 ViT-B/14 with two-stage fine-tuning and 16-pass test-time augmentation
- Ranked 27th on Kaggle (score 0.836), raising validation accuracy from ∼63% to 82.79%

DAWLab – Collaborative Version Control for Music | *React, TypeScript, Electron* — Jan 2026 – Jun 2026

- Designed a Git-inspired version control system for DAW files with branching, merging, and rollback
- Built a full-stack Electron app: React 18 + TypeScript, FastAPI, Supabase, Clerk, and Sentry
- Adopted by a UCSC music production course; supports Ableton Live, Logic Pro X, and FL Studio

LLM-Powered MITRE ATT&CK Command Generator | *Python, HuggingFace, RAG* — Sep 2025 – Jan 2026

- Built a multi-agent AI pipeline for adversarial command generation aligned with the MITRE ATT&CK framework
- Fine-tuned Llama 3.1 and Qwen via HuggingFace Transformers with a RAG grounding pipeline
- Spans 193+ ATT&CK techniques across 14 tactic categories, reducing hallucinated commands

Slug Board – UCSC Campus Events Platform | *React.js, Tailwind CSS, SQL, Python* — Jun 2025 – Aug 2025

- Led product as Product Owner for a 5-person team building UCSC's campus event discovery platform
- Bridged Google OAuth with a Supabase profile system and a Python event-scraping pipeline
- Shipped the home feed, event creation, and profile editing using Scrum-based agile development

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Bash
ML/AI: PyTorch, HuggingFace, CLAP, DINOv2, RL (PPO, DDPG, SAC), RAG, scikit-learn, pandas, NumPy
Cybersecurity: MITRE ATT&CK, Wireshark, nmap, hashcat, Kali Linux, Exploitation, Reverse Engineering, Cryptography, Network Security, CTF Competitions
Infrastructure/Web: Kubernetes, Docker, Nautilus HPC, CARLA, Git, CI/CD, React 18, TypeScript, Vite, Electron, Tailwind CSS, FastAPI, Supabase, Firebase, Clerk
Languages Spoken: English (Native), Japanese (Native/Bilingual), Spanish (Basic)